J-space, Part 2: The Model Matrix

From lab result to publishable finding — living handout

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Campaign started 2026-07-27 — preregistered before data (`27c6d3c`)

1 Where Part 1 left off, and what Part 2 must decide

Part 1 (Lab 37) replicated the descriptive geometry of the Anthropic workspace paper on `Olmo-3-32B-Think` at ∼10× thinner capacity, found the paper's static causal dissociation *null* on both OLMo and `Qwen3.6-27B` under energy-matched instruments, established per-item frozen J-ablation as a control-clean cross-model causal handle on retrieved content (−2.9/−2.4 nats), and measured a foil-calibrated 46-step workspace-ahead-of-text lead. Part 2 adjudicates *why the dissociation is missing*:

	hypothesis	discriminator
H1	externalization (think-training moved the workspace into tokens)	Workstream A: matched-pretraining sibling `Olmo-3.1-32B-Instruct`, Qwen matrix completion, `gemma-4-31B-it`
H2	occupancy (live J-space ≈ the output stream itself)	Workstream D: output-occupancy index across the matrix
H3	training-lab specifics	unfalsifiable; shrunk by exhausting the rest
H5	residual instruments	B (frozen-logit, fit-size, Dolma corpus, dose/persistence selection), C (load-demanding tasks, paper's own sets)

Standards: $n \geq 60$ headline cells, two seeds on decisive grids, bootstrap CIs, BH-FDR across the matrix, preregistered directional predictions (`preregistration.md`).

2 Workstream A

2.1 A0 — does the J-dictionary survive post-training?

The part-1 Think lens, applied unchanged (donor J, recipient unembedding) to `Olmo-3.1-32B-Instruct` — same base model, different post-training. Preregistered gate: probes ≥15/21 and multihop pass@1 $\geq 0.85\times$ donor $= 0.2408$.

Verdict: `TRANSFER_FAIL`, informative. The readout basis is static across post-training (unembed row cosine median 0.9984; final-norm gain cosine 1.000), direct-fact probes transfer exactly (17/21 both), the shortlist transfers — what post-training blunted is specifically the *rank-1 sharpening*, the very component that was the J-lens's distinctive advantage on Think (+41% relative pass@1 there, +8.5% here). The prereg FAIL rule fires: fit the sibling's own 120-prompt lens. Recovery to ∼0.28 ⇒ mid-band J-map drift under post-training; no recovery ⇒ Instruct resolves the silent bridge more weakly — itself an H1-flavored datum.

*Draft campaign handout; grows a subsection per banked phase. Part-1 handout: `jspace/handout/olmo32b_jspace_handout.pdf`.

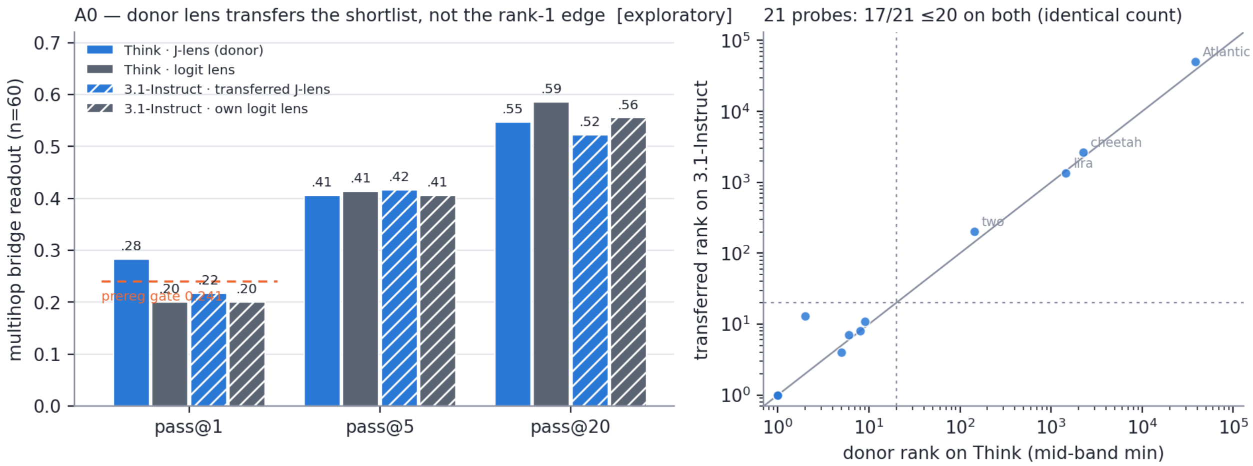

Figure 1: **A0 transfer gate.** Left: the transferred lens reads the bridge-entity *shortlist* at donor level (pass@5 0.417, pass@20 0.522 vs donor 0.41/0.55) but loses the rank-1 edge (0.217 vs 0.283, below the preregistered floor, dashed). Right: per-probe mid-band ranks — 17/21 $\leq$20 on *both* models, same items missing.

3 Workstream B

Reproducibility

Run dir `special-lab-1/part2_20260727/` (metrics per model slug, lenses, logs); code + results mirrored to `interpretability/jspace/part2/` on branch `interp_jspace_part2`; every figure regenerates from metrics via `scripts/p2fig.py`; master dataset `report/matrix_master.csv`.